

Chapter 8

Elements of realization theory

The purpose of this chapter is to discuss some concepts and to present some algorithms, related to the problem of determining a state space model compatible with the specifications of its input/output behavior. The solution of this problem is based on a set of methodologies that are commonly referred to as *realization theory*. State representations allow to develop efficient procedures for the design of a regulator, the construction of optimal control and filtering laws, and, in general, the synthesis of algorithms for real time data processing. On the other hand, quite frequently system specifications and quality indices refer to the input/output map, so that the problem of associating to the external representation a state model that “realizes” it arises. In other words we aim at describing, in terms of “internal” structure and functioning, a device whose behavior at the input and output gates has been assigned.

The results of the realization theory are also relevant in areas different from control, such as, for example, automata theory, the synthesis of linear electrical networks, and recursive algorithms. It is intuitive that they are of considerable importance also for the identification problem, in which we try to single out, with recursive techniques, a state model that accounts for the behavior exhibited by a given system.

Throughout the chapter we will just consider input/output maps that can be realized by linear and time-invariant state space models.

8.1 Realization of linear input-output maps

Let $\Sigma = (F, G, H, D)$ be a discrete time system in state space form, with m inputs and p outputs, and assume that its initial state is zero. As we have shown in Chapter 2, the relation (the “map”) between (the formal power series representations of) its inputs and outputs is expressed by the transfer matrix of Σ

$$W_{\Sigma}(z) = H(zI - F)^{-1}G + D, \quad (8.1)$$

in the following form

$$Y(z) = W_{\Sigma}(z)U(z). \quad (8.2)$$

The issue we intend to discuss now can be viewed as the inverse problem: given a linear input/output map expressed in the form

$$Y(z) = W(z)U(z), \quad (8.3)$$

where $W(z)$ is a $p \times m$ matrix of rational functions, we plan to obtain (the matrices F, G, H, D of) a linear dynamical system in state space form Σ , such that the transfer matrix of Σ coincides with $W(z)$. Any system Σ in state space form that solves this problem, namely that satisfies the equation $W_\Sigma(z) = W(z)$, is called a *realization* of $W(z)$.

Remark 1 The input /(forced) output map of a linear discrete time system $\Sigma = (F, G, H, D)$ can be alternatively represented by an ARMA model. In fact, with reference to a strictly proper system ($D = 0$) in section. 2.3.3 we derived the relation

$$\sum_{i=0}^n \alpha_i \mathbf{y}(t+i) = \sum_{i=0}^{n-1} H N_i G \mathbf{u}(t+i)$$

and the extension to the case $D \neq 0$ is easy:

$$\sum_{i=0}^n \alpha_i \mathbf{y}(t+i) = D \mathbf{u}(t+n) + \sum_{i=0}^{n-1} (H N_i G + \alpha_i D) \mathbf{u}(t+i). \quad (8.4)$$

Viceversa, an ARMA model written in the form

$$P(\sigma^{-1}) \mathbf{y}(t) = Q(\sigma^{-1}) \mathbf{u}(t) \quad (8.5)$$

induces an input/output map given by the transfer matrix

$$W(z) = P(z^{-1})^{-1} Q(z^{-1}), \quad (8.6)$$

and, consequently, the problem of realizing the ARMA model (8.5) by means of a state model is reduced to the realization of the rational matrix (8.6).

Since all the entries of (8.1) are proper rational functions, it is obvious that a rational matrix that includes non proper elements cannot be realized by a dynamical system.

Moreover, we can always convert the problem of realizing a proper rational matrix into the realization of a strictly proper rational matrix. Indeed, a $p \times m$ proper rational matrix $W(z)$ can be decomposed (in a unique way) in the following form $W(z) = W_{sp}(z) + D$, where $W_{sp}(z)$ is strictly proper rational and D is a $p \times m$ constant matrix. To that purpose, one applies the euclidean division algorithm to the numerator and the denominator of each rational entry $w_{ij}(z) = b_{ij}(z)/a_{ij}(z)$ of $W(z)$ and rewrite the numerator in the form

$$b_{ij}(z) = d_{ij} a_{ij}(z) + r_{ij}(z), \quad d_{ij} \in \mathbb{R}, \quad \deg r_{ij}(z) < \deg a_{ij}(z)$$

Consequently, each rational entry, as well as the transfer matrix, splits into the sum of two addends, namely

$$w_{ij}(z) = \frac{r_{ij}(z)}{a_{ij}(z)} + d_{ij}$$

and

$$W(z) = \left[\frac{r_{ij}(z)}{a_{ij}(z)} \right] + [d_{ij}] = W_{sp}(z) + D.$$

Once a realization $\Sigma_{sp} = (F, G, H)$ of $W_{sp}(z)$ has been constructed, a realization of $W(z)$ is given by $\Sigma = (F, G, H, D)$.

Proposition 8.1.1 [REALIZATION - DISCRETE TIME CASE] *Given any strictly proper rational matrix $W(z)$ of dimension $p \times m$, there exists a linear dynamical system that provides a realization of $W(z)$.*

We prove this statement by providing a set of constructive procedures to obtain a realization of $W(z)$. Single input/single output systems will be considered first.

PROOF *Scalar case* ($m = p = 1$).

Consider a strictly proper rational function, expressed as the ratio of two polynomials $a(z)$ and $b(z)$, with $b(z)$ monic denominator

$$W(z) = \frac{b(z)}{a(z)} = \frac{\beta_0 + \beta_1 z + \beta_2 z^2 + \dots + \beta_{n-1} z^{n-1}}{\alpha_0 + \alpha_1 z + \alpha_2 z^2 + \dots + z^n}. \quad (8.7)$$

Direct computation shows that the system in controllable canonical form $\Sigma_c = (F_c, \mathbf{g}_c, H_c)$

$$\begin{aligned} F_c &= \begin{bmatrix} 0 & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ -\alpha_0 & -\alpha_1 & -\alpha_2 & \dots & -\alpha_{n-1} \end{bmatrix}, \quad \mathbf{g}_c = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \\ H_c &= [\beta_0 \quad \beta_1 \quad \beta_2 \quad \dots \quad \beta_{n-1}] \end{aligned} \quad (8.8)$$

and the system in observable canonical form $\Sigma_o = (F_o, \mathbf{g}_o, H_o)$

$$\begin{aligned} F_o &= \begin{bmatrix} 0 & & & & -\alpha_0 \\ 1 & & & & -\alpha_1 \\ & 1 & & & -\alpha_2 \\ & & \ddots & & \vdots \\ & & & 1 & -\alpha_{n-1} \end{bmatrix}, \quad \mathbf{g}_o = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \end{bmatrix} \\ H_o &= [0 \quad 0 \quad \dots \quad 0 \quad 1] \end{aligned} \quad (8.9)$$

are both realizations of $W(z)$, reachable the former, observable the latter.

Indeed, by relying on the arguments proposed in section 6.3, we derive the equality

$$H_c(zI - F_c)^{-1} \mathbf{g}_c = \frac{\beta_0 + \beta_1 z + \beta_2 z^2 + \dots + \beta_{n-1} z^{n-1}}{\alpha_0 + \alpha_1 z + \alpha_2 z^2 + \dots + z^n}.$$

This proves that Σ_c realizes $W(z)$. On the other hand the transfer function of the single input/single output systems Σ_o and Σ_c is the same, as they are one the dual of the other. ■

We remark that (8.8) is reachable and (8.9) is observable regardless of whether the representation selected for $W(z)$ is irreducible (i.e., devoid of non trivial common factors between the numerator and the denominator). Denote by

$$\frac{b_m(z)}{a_m(z)} = W(z)$$

the unique irreducible representation of $W(z)$ with monic denominator, so that all the representations of $W(z)$ with monic denominator have the following form

$$W(z) = \frac{b_m(z)c(z)}{a_m(z)c(z)}, \quad (8.10)$$

where $c(z)$ is an arbitrary monic polynomial. Once a specific representation of $W(z)$ has been selected in (8.10), systems (8.8) and (8.9) provide a reachable and an observable realization, respectively. The dimensions of these realizations depend on the degree of $c(z)$, and realizations that correspond to different choices of $c(z)$ cannot be algebraically equivalent, as their state updating matrices exhibit different characteristic polynomials.

PROOF OF 8.1.1 (CONTINUED) *Matrix case* ($m > 1$ or $p > 1$).

Given a $p \times m$ matrix of strictly proper rational functions

$$W(z) = \left[\frac{b_{ij}(z)}{a_{ij}(z)} \right]_{\substack{i=1,2,\dots,p \\ j=1,2,\dots,m}},$$

let

$$a(z) = \alpha_0 + \alpha_1 z + \dots + z^\nu,$$

be the monic least common multiple (l.c.m.) of the denominators $a_{ij}(z)$. Then $a(z)W(z)$ is a polynomial matrix and can be rewritten in the form

$$a(z)W(z) = B_0 + B_1 z + \dots + B_{\nu-1} z^{\nu-1}, \quad (8.11)$$

where the matrices B_i belong to $\mathbb{R}^{p \times m}$. By direct computation (see Exercise 8.1.1 below) one checks that the reachable system of dimension $m\nu$

$$\begin{aligned} F_c &= \begin{bmatrix} 0_m & I_m & & & \\ & & I_m & & \\ & & & \ddots & \\ & & & & I_m \\ -\alpha_0 I_m & -\alpha_1 I_m & -\alpha_2 I_m & \dots & -\alpha_{\nu-1} I_m \end{bmatrix}, \quad G_c = \begin{bmatrix} 0_m \\ 0_m \\ \vdots \\ 0_m \\ I_m \end{bmatrix} \\ H_c &= \begin{bmatrix} B_0 & B_1 & B_2 & \dots & B_{\nu-1} \end{bmatrix} \end{aligned} \quad (8.12)$$

and the observable system of dimension $p\nu$

$$\begin{aligned} F_o &= \begin{bmatrix} 0_p & & & -\alpha_0 I_p \\ I_p & & & -\alpha_1 I_p \\ & I_p & & -\alpha_2 I_p \\ & & \ddots & \vdots \\ & & & I_p & -\alpha_{\nu-1} I_p \end{bmatrix}, \quad G_o = \begin{bmatrix} B_0 \\ B_1 \\ B_2 \\ \vdots \\ B_{\nu-1} \end{bmatrix} \\ H_o &= \begin{bmatrix} 0_p & 0_p & \dots & 0_p & I_p \end{bmatrix} \end{aligned} \quad (8.13)$$

provide two state space realizations of $W(z)$. ■

Taking into account that the rational functions in $W(z)$ admit infinitely many representations, it is clear that also in the multi-input/multi-output case it is possible to construct

realizations (8.12) and (8.13) of $W(z)$ of arbitrarily high dimension. Realizations (8.12) and (8.13), which reduce to (8.8) and (8.9), respectively, when $m = p = 1$, show that any strictly proper rational matrix is realizable by means of a linear time invariant linear system.

- EXERCISE 8.1.1 Prove that matrices F_c, G_c, H_c in (8.12) provide a reachable realization of $W(z)$.
 ‡ *Solution:* (i) Rewrite all the entries $b_{ij}(z)/a_{ij}(z)$ of $W(z)$ with a unique monic denominator $a(z)$, thus obtaining $w_{ij}(z) = \frac{\bar{b}_{ij}(z)}{a(z)}$, $i = 1, 2, \dots, p$, $j = 1, 2, \dots, m$. The coefficients of the polynomials $\bar{b}_{ij}(z) = \beta_0^{(ij)} + \beta_1^{(ij)}z + \dots + \beta_{\nu-1}^{(ij)}z^{\nu-1}$ and $a(z)$ allow to get reachable realizations (each one with dimension ν), of the scalar rational functions $w_{ij}(z)$

$$\begin{aligned} \Sigma^{(i,j)} &= (F_c, \mathbf{g}_c, H_c^{(i,j)}) \\ &= \left(\begin{bmatrix} 0 & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ -\alpha_0 & -\alpha_1 & -\alpha_2 & \dots & -\alpha_{\nu-1} \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, H_c^{(i,j)} = [\beta_0^{(i,j)} \quad \beta_1^{(i,j)} \quad \beta_2^{(i,j)} \quad \dots \quad \beta_{\nu-1}^{(i,j)}] \right) \end{aligned}$$

ii) Consider the system $\Sigma = (F, G, H)$, of dimension $m\nu$, with

$$\begin{aligned} zI - F &= \begin{bmatrix} zI_\nu - F_c & & & \\ & zI_\nu - F_c & & \\ & & \ddots & \\ & & & zI_\nu - F_c \end{bmatrix}, \quad G = \begin{bmatrix} \mathbf{e}_\nu & & & \\ & \mathbf{e}_\nu & & \\ & & \ddots & \\ & & & \mathbf{e}_\nu \end{bmatrix}, \\ H &= \begin{bmatrix} \beta_0^{(11)} & \beta_1^{(11)} & \dots & \beta_{\nu-1}^{(11)} & \dots & \beta_0^{(1m)} & \beta_1^{(1m)} & \dots & \beta_{\nu-1}^{(1m)} \\ \beta_0^{(21)} & \beta_1^{(21)} & \dots & \beta_{\nu-1}^{(21)} & \dots & \beta_0^{(2m)} & \beta_1^{(2m)} & \dots & \beta_{\nu-1}^{(2m)} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \beta_0^{(p1)} & \beta_1^{(p1)} & \dots & \beta_{\nu-1}^{(p1)} & \dots & \beta_0^{(pm)} & \beta_1^{(pm)} & \dots & \beta_{\nu-1}^{(pm)} \end{bmatrix}. \end{aligned}$$

The (i, j) -th entry of its transfer matrix $H(zI - F)^{-1}G$ can be obtained as the product of the i -th row of H , the matrix $(zI - F)^{-1}$ and the j -column of G , in the order:

$$\begin{aligned} & [\beta_0^{(i1)} \quad \beta_1^{(i1)} \quad \dots \quad \dots \quad \dots \quad \beta_{\nu-1}^{(im)}] \begin{bmatrix} (zI_\nu - F_c)^{-1} & & & \\ & (zI_\nu - F_c)^{-1} & & \\ & & \ddots & \\ & & & (zI_\nu - F_c)^{-1} \end{bmatrix} \text{col}_j G, \\ &= \begin{bmatrix} \beta_0^{(i1)} & \dots & \beta_{\nu-1}^{(i1)} & \dots & \underbrace{\beta_0^{(ij)} \quad \dots \quad \beta_{\nu-1}^{(ij)}}_{j\text{-th block}} & \dots & \beta_0^{(im)} & \dots & \beta_{\nu-1}^{(im)} \end{bmatrix} \begin{bmatrix} \mathbf{0}_{\nu \times \nu} \\ \vdots \\ (zI_\nu - F_c)^{-1} \mathbf{e}_\nu \\ \vdots \\ \mathbf{0}_{\nu \times \nu} \end{bmatrix} \leftarrow j\text{-th block} \\ &= \frac{\bar{b}_{ij}(z)}{a(z)}. \end{aligned}$$

Hence $\Sigma = (F, G, H)$ realizes $W(z)$. The pair (F, G) is obviously reachable.

(iii) One obtains the system (F_c, G_c, H_c) from (F, G, H) by resorting to a “cogredience transformation”: there exists a permutation matrix Π such that $F_c = \Pi^T F \Pi$, $G_c = \Pi^T G$, $H_c = H \Pi$. Determine the permutation one has to apply to the columns and the rows of F .

Remark 2 [REALIZATION - CONTINUOUS TIME CASE] The transfer matrix of a continuous time system $\Sigma = (F, G, H)$, namely the proper rational matrix $W(s) = H(sI - F)^{-1}G$, coincides, except for the name of the variable, with the transfer matrix of the discrete

time system characterized by the same matrices F, G, H . Consequently the realization problem for the input/output map of a continuous time system, assigned by means of a rational matrix $W(s)$, can be solved by resorting to the methods and algorithms proposed in the discrete time case.

8.2 Minimal realizations

As we have seen in Section 8.1, if a rational matrix $W(z)$ is realizable, it admits infinitely many state space realizations, of arbitrarily high dimension. The purpose of this section is to analyze the properties of the realizations of $W(z)$ whose dimension is as small as possible. Throughout the chapter, unless otherwise stated, we assume that $W(z)$ is strictly proper.

Definition 8.2.1 *A realization $\Sigma = (F, G, H)$ of $W(z)$ is called “minimal” if for any realization $\Sigma' = (F', G', H')$ of $W(z)$ one has*

$$\dim \Sigma \leq \dim \Sigma'. \quad (8.14)$$

The following proposition presents the fundamental property of the minimal realizations.

Proposition 8.2.2 [MINIMALITY, REACHABILITY AND OBSERVABILITY] *A realization $\Sigma = (F, G, H)$ of the rational matrix $W(z)$ is minimal if and only if it is reachable and observable.*

PROOF Let Σ be a minimal realization of $W(z)$ and suppose, by contradiction, that Σ is not reachable. By a suitable change of basis in the state space, we put the system in standard reachability form

$$\bar{\Sigma} = (T^{-1}FT, T^{-1}G, HT) = (\bar{F}, \bar{G}, \bar{H}) = \left(\begin{bmatrix} \bar{F}_{11} & \bar{F}_{12} \\ 0 & \bar{F}_{22} \end{bmatrix}, \begin{bmatrix} \bar{G}_1 \\ 0 \end{bmatrix}, [\bar{H}_1 \quad \bar{H}_2] \right)$$

and consider the reachable subsystem of $\bar{\Sigma}$

$$\bar{\Sigma}^R = (\bar{F}_{11}, \bar{G}_1, \bar{H}_1).$$

As $W(z)$ is the transfer matrix of Σ , it is also the transfer matrix of the algebraically equivalent system $\bar{\Sigma}$, and hence of its reachable subsystem $\bar{\Sigma}^R$. However, $\bar{\Sigma}^R$ has dimension smaller than Σ , which contradicts the assumption that Σ is a minimal realization of $W(z)$. The proof that minimality implies observability follows the same lines.

What we have proved hitherto shows that the set of minimal realizations of $W(z)$ is included in the set of its reachable and observable realizations. The proof will be complete once one has shown that the two sets coincide.

To that purpose we need the following lemma.

Lemma 8.2.3 *All reachable and observable realizations of the strictly proper rational matrix $W(z)$ have the same dimension.*

PROOF The matrix $W(z)$ can be expressed as a formal power series in the indeterminate z^{-1} :

$$W(z) = M_1 z^{-1} + M_2 z^{-2} + \dots + M_k z^{-k} + \dots \quad (8.15)$$

The constant matrices $M_k \in \mathbb{R}^{p \times m}$ are the so called *Markov coefficients* of $W(z)$ and the sequence $M_1, M_2, M_3, \dots$ or, equivalently, the associated power series (8.15), is the *impulse response* of the system: in fact, with the same reasoning as in Section 2.1, we can check that the sequence of the j -th columns of the matrices M_k is the forced output one gets when a (Kronecker) impulse is applied to the j -th input. The decomposition procedure of a proper rational matrix into a constant and a strictly proper rational part (see Section 8.1) allows to recursively compute the Markov coefficients of $W(z)$:

$$zW(z) = M_1 + W^{(1)}(z), \quad zW^{(1)}(z) = M_2 + W^{(2)}(z), \quad \dots, \quad zW^{(k)}(z) = M_{k+1} + W^{(k+1)}(z), \dots$$

Clearly $W^{(k)}(z)$ are strictly proper rational matrices, $k = 1, 2, \dots$.

Given an arbitrary realization $\Sigma = (F, G, H)$ of $W(z)$, the rational matrix can be expressed as follows

$$\begin{aligned} W(z) &= H(zI - F)^{-1}G \\ &= Hz^{-1}(I - Fz^{-1})^{-1}G \\ &= HGz^{-1} + HFGz^{-2} + \dots + HF^k Gz^{-k-1} + \dots \end{aligned} \quad (8.16)$$

Comparing (8.15) with (8.16) we deduce that, the matrices F, G, H of any realization $\Sigma = (F, G, H)$ of $W(z)$ are related to the Markov coefficients of $W(z)$ by the following relations

$$M_1 = HG, \quad M_2 = HFG, \quad \dots, \quad M_{k+1} = HF^k G, \quad \dots \quad (8.17)$$

Suppose now that both $\Sigma = (F, G, H)$ and $\bar{\Sigma} = (\bar{F}, \bar{G}, \bar{H})$ are reachable and observable realizations of $W(z)$, and assume, by contradiction, that Σ has dimension n and $\bar{\Sigma}$ is minimal and has dimension $\bar{n} < n$.

Denote by $\mathcal{R}$ and $\mathcal{O}$, respectively, the reachability and observability matrices of Σ . Each one of them has rank n . Moreover, denote by $\bar{\mathcal{R}}_n$ and $\bar{\mathcal{O}}_n$ the n -steps reachability and observability matrices of $\bar{\Sigma}$. Clearly they have rank $\bar{n}$.

The explicit structure of the products $\mathcal{O}\mathcal{R}$ and $\bar{\mathcal{O}}_n\bar{\mathcal{R}}_n$ is given by

$$\mathcal{O}\mathcal{R} = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{n-1} \end{bmatrix} [G \quad FG \quad \dots \quad F^{n-1}G] = \begin{bmatrix} HG & HFG & \dots & HF^{n-1}G \\ HFG & HF^2G & \dots & HF^nG \\ \dots & \dots & \dots & \dots \\ HF^{n-1}G & HF^nG & \dots & HF^{2n-2}G \end{bmatrix} \quad (8.18)$$

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$$\bar{\mathcal{O}}_n\bar{\mathcal{R}}_n = \begin{bmatrix} \bar{H} \\ \bar{H}\bar{F} \\ \vdots \\ \bar{H}\bar{F}^{n-1} \end{bmatrix} [\bar{G} \quad \bar{F}\bar{G} \quad \dots \quad \bar{F}^{n-1}\bar{G}] = \begin{bmatrix} \bar{H}\bar{G} & \bar{H}\bar{F}\bar{G} & \dots & \bar{H}\bar{F}^{n-1}\bar{G} \\ \bar{H}\bar{F}\bar{G} & \bar{H}\bar{F}^2\bar{G} & \dots & \bar{H}\bar{F}^n\bar{G} \\ \dots & \dots & \dots & \dots \\ \bar{H}\bar{F}^{n-1}\bar{G} & \bar{H}\bar{F}^n\bar{G} & \dots & \bar{H}\bar{F}^{2n-2}\bar{G} \end{bmatrix} \quad (8.19)$$

Since both Σ and $\bar{\Sigma}$ realize $W(z)$, (8.17) implies

$$M_{k+1} = \bar{H}\bar{F}^k\bar{G} = HF^kG, \quad \forall k \geq 0.$$

Taking into account (8.18) and (8.19), we get the straightforward equality

$$\mathcal{OR} = \bar{\mathcal{O}}_n \bar{\mathcal{R}}_n.$$

The rank of the matrix product $\mathcal{OR}$ does not exceed n , as both factors have rank n . As a matter of fact, the rank of the product is exactly n , as we shall prove by singling out n linearly independent vectors in the image of $\mathcal{OR}$.

Note first that, since $\mathcal{R}$ has rank n , there exist n canonical vectors¹ $\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_n} \in \mathbb{R}^{nm}$ with the property that

$$\mathcal{R}\mathbf{e}_{i_1}, \mathcal{R}\mathbf{e}_{i_2}, \dots, \mathcal{R}\mathbf{e}_{i_n} \quad (8.20)$$

are linearly independent vectors² in $\mathbb{R}^n$. Next, we show that the vectors $\mathcal{OR}\mathbf{e}_{i_1}, \mathcal{OR}\mathbf{e}_{i_2}, \dots, \mathcal{OR}\mathbf{e}_{i_n}$, are linearly independent, too. In fact, suppose we are given an linear combination of $\mathcal{OR}\mathbf{e}_{i_\nu}$, $\nu = 1, \dots, n$, that sums up to zero:

$$\mathbf{0} = \sum_{\nu=1}^n \alpha_\nu (\mathcal{OR}\mathbf{e}_{i_\nu}) = \mathcal{O} \left(\sum_{\nu=1}^n \alpha_\nu \mathcal{R}\mathbf{e}_{i_\nu} \right)$$

The observability of Σ implies $\sum_{\nu=1}^n \alpha_\nu \mathcal{R}\mathbf{e}_{i_\nu} = \mathbf{0}$ and the independence of the vectors (8.20) implies $\alpha_\nu = 0, \forall \nu$ and therefore the independence of $\mathcal{OR}\mathbf{e}_{i_\nu}$, $\nu = 1, \dots, n$,

From $\mathcal{OR} = \bar{\mathcal{O}}_n \bar{\mathcal{R}}_n$ one gets

$$n = \text{rank}(\mathcal{OR}) = \text{rango}(\bar{\mathcal{O}}_n \bar{\mathcal{R}}_n),$$

which is impossible, because $\text{rank}(\bar{\mathcal{O}}_n \bar{\mathcal{R}}_n)$ cannot exceed $\bar{n}$.

This proves the lemma, and concludes the proof of Proposition 8.2.2. ■

Given an arbitrary realization $\Sigma = (F, G, H)$ of a strictly proper rational matrix $W(z)$, it is easy to construct a minimal (i.e. reachable and observable) realization: it will suffice to compute the observable subsystem Σ_O of Σ and, as a second step, the reachable subsystem Σ_{OR} of Σ_O (note that Σ_{OR} preserves the observability property of Σ_O).

Therefore, a general procedure for obtaining a minimal realization of any strictly proper rational matrix $W(z)$ consists in complementing the realization algorithms described in Section 8.1, that provide a reachable or an observable realization Σ of $W(z)$, with suitable reduction procedures that eliminate its unobservable and/or unreachable subsystem(s).

At this point a natural question arises as to whether some connection exists between the triples of matrices of two arbitrary minimal realizations of $W(z)$, whatever the procedure used to obtain them were. As the following proposition shows, all minimal realizations of $W(z)$ are algebraically equivalent, i.e., essentially reducible to a unique reachable and observable dynamical system, represented with respect to different bases of the state space.

¹Equivalently, there exist n inputs, each one consisting of a unitary sample, that produce independent states at suitable instants of the time interval $[0, n-1]$.

²I.e. linearly independent states the system reaches.

Proposition 8.2.4 [ALGEBRAIC EQUIVALENCE OF MINIMAL REALIZATIONS] *Let $\Sigma_1 = (F_1, G_1, H_1)$ and $\Sigma_2 = (F_2, G_2, H_2)$ be minimal realizations of $W(z)$ and let n be their dimension. Then there exists a nonsingular matrix $T \in \mathbb{R}^{n \times n}$ such that*

$$F_1 = T^{-1}F_2T, \quad G_1 = T^{-1}G_2, \quad H_1 = H_2T. \quad (8.21)$$

PROOF Let $\mathcal{R}_1, \mathcal{R}_2, \mathcal{O}_1$ and $\mathcal{O}_2$ denote the reachability and observability matrices of Σ_1 and Σ_2 , respectively. By the proof of Lemma 8.2.3 we have

$$\mathcal{O}_1\mathcal{R}_1 = \mathcal{O}_2\mathcal{R}_2, \quad (8.22)$$

whence it follows

$$\mathcal{O}_2^T \mathcal{O}_1 \mathcal{R}_1 = \mathcal{O}_2^T \mathcal{O}_2 \mathcal{R}_2. \quad (8.23)$$

The rank of the matrix product $\mathcal{O}_2^T \mathcal{O}_2 \in \mathbb{R}^{n \times n}$ is n , as it coincides with the rank of $\mathcal{O}_2$. Therefore $\mathcal{O}_2^T \mathcal{O}_2$ is invertible, and (8.23) allows to single out $\mathcal{R}_2$

$$\left(\mathcal{O}_2^T \mathcal{O}_2\right)^{-1} \mathcal{O}_2^T \mathcal{O}_1 \mathcal{R}_1 = \mathcal{R}_2. \quad (8.24)$$

We are now in a position to show that the matrix T defined as

$$T := \left(\mathcal{O}_2^T \mathcal{O}_2\right)^{-1} \mathcal{O}_2^T \mathcal{O}_1 \quad (8.25)$$

satisfies all the relations (8.21).

i) We preliminarily remark that (8.24) and (8.25) imply

$$T\mathcal{R}_1 = \mathcal{R}_2, \quad (8.26)$$

hence the rank of T is greater than or equal to the rank of $\mathcal{R}_2$ (which is n , as Σ_2 is reachable) and consequently T is an invertible matrix.

ii) From (8.26) one immediately gets $TG_1 = G_2$.

iii) From (8.22) and (8.26) one gets $\mathcal{O}_2\mathcal{R}_2 = \mathcal{O}_1T^{-1}\mathcal{R}_2$, whence $(\mathcal{O}_2 - \mathcal{O}_1T^{-1})(\mathcal{R}_2\mathcal{R}_2^T) = 0$. Since $\mathcal{R}_2\mathcal{R}_2^T$ is invertible, we have

$$\mathcal{O}_2 = \mathcal{O}_1T^{-1} \quad (8.27)$$

and finally $H_2 = H_1T^{-1}$.

iv) Using the same procedure as in the proof of Lemma 8.2.3, it is easy to check the following identity

$$\mathcal{O}_1F_1\mathcal{R}_1 = \begin{bmatrix} M_2 & M_3 & \dots & M_{n+1} \\ M_3 & M_4 & \dots & M_{n+2} \\ & \dots & \dots & \\ M_{n+1} & M_{n+2} & \dots & M_{2n} \end{bmatrix} = \mathcal{O}_2F_2\mathcal{R}_2. \quad (8.28)$$

Keeping into account (8.26) and (8.27), the last member of (8.28) becomes

$$\mathcal{O}_1 F_1 \mathcal{R}_1 = \mathcal{O}_2 F_2 \mathcal{R}_2 = \mathcal{O}_1 T^{-1} F_2 T \mathcal{R}_1.$$

We therefore have

$$0 = \mathcal{O}_1 (F_1 - T^{-1} F_2 T) \mathcal{R}_1 = (\mathcal{O}_1^T \mathcal{O}_1) (F_1 - T^{-1} F_2 T) (\mathcal{R}_1 \mathcal{R}_1^T) \quad (8.29)$$

and the invertibility of both $\mathcal{O}_1^T \mathcal{O}_1$ and $\mathcal{R}_1 \mathcal{R}_1^T$ implies $F_1 - T^{-1} F_2 T = 0$. $\blacksquare$

- EXERCISE 8.2.1 Prove the uniqueness of the matrix T that induces the algebraic equivalence in (8.21).

To conclude this section, we mention that it is possible to construct a minimal realization of a strictly proper rational matrix $W(z)$ directly, thus avoiding the reduction procedure we should resort to when the realization algorithm leads to a unobservable and/or unreachable subsystem. We confine ourselves to single input/single output systems ($m = p = 1$), as the multi-input/multi-output case is rather difficult. The construction procedure is a direct consequence of the following proposition.

Proposition 8.2.5 [MINIMAL REALIZATIONS OF STRICTLY PROPER RATIONAL FUNCTIONS] *Let $\frac{\bar{b}(z)}{\bar{a}(z)}$ be an irreducible representation of the strictly proper rational function $w(z)$, where $\bar{n}$ is the degree $\bar{a}(z)$, and let $\Sigma = (F, \mathbf{g}, H)$ be a realization of $w(z)$. The following are equivalent:*

- i) Σ is a minimal realization of $w(z)$,
- ii) $\dim(\Sigma) = \bar{n}$,
- iii) the polynomials $H \text{adj}(zI - F) \mathbf{g}$ and $\det(zI - F)$ are coprime.

PROOF (i) $\Leftrightarrow$ (ii) The algorithm presented in Section 8.1 shows that there exists a realization of $w(z)$ whose dimension equals the degree of the denominator polynomial $\bar{a}(z)$. Therefore, if $\Sigma = (F, \mathbf{g}, H)$ is a minimal realization, its dimension cannot exceed $\bar{n}$. On the other hand, it cannot be smaller than $\bar{n}$, otherwise the denominator of the representation of $w(z)$ given by

$$w(z) = \frac{H \text{adj}(zI - F) \mathbf{g}}{\det(zI - F)} \quad (8.30)$$

would exhibit a degree smaller than $\bar{a}(z)$. This is impossible as $\bar{b}(z)/\bar{a}(z)$ is an irreducible representation.

(iii) $\Leftrightarrow$ (i) If $H \text{adj}(zI - F) \mathbf{g}$ and $\det(zI - F)$ are coprime, their quotient provides an irreducible representation of $w(z)$ and, by the above equivalence, the dimension of a minimal realization coincides with the degree of $\det(zI - F)$. Consequently $\Sigma = (F, \mathbf{g}, H)$ is a minimal realization.

If they are not coprime, canceling out all common factors in (8.30) leads to a representation of $w(z)$ whose denominator has degree $\bar{n}$, smaller than $\deg \det(zI - F) = \dim \Sigma$. Therefore there exists an $\bar{n}$ -dimensional realization of $w(z)$ and Σ fails to be minimal. $\blacksquare$

8.3 Connections between internal and external stability

A linear system $\Sigma = (F, G, H)$ is called *externally stable*, or *BIBO³ stable* if every input $\mathbf{u}(\cdot)$ with bounded norm (i.e. $\sup_{t \geq 0} \|\mathbf{u}(t)\| < \infty$) induces a *forced* output $\mathbf{y}_f(\cdot)$ with bounded norm (i.e. $\sup_{t \geq 0} \|\mathbf{y}_f(t)\| < \infty$)⁴.

As the external stability is a property of the input/output map (with zero initial conditions) it only depends on the structure of the transfer matrix of Σ , that in turn will be called (BIBO) stable. Consequently, as far as the external stability is concerned, any two realizations of the same transfer matrix exhibit the same stability character: either they are both externally stable or they are not.

Remark It is useful to underline that the external stability definition exclusively refers to the behaviour of the *forced* output of a system. Let us consider the following example, that illustrates how the unforced component could possibly jeopardize the output boundedness of a BIBO stable system

$$\begin{aligned} \mathbf{x}(t+1) &= \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t) = F\mathbf{x}(t) + \mathbf{g}u(t) \\ y(t) &= \begin{bmatrix} 1 & 1 \end{bmatrix} \mathbf{x}(t) = H\mathbf{x}(t) \end{aligned}$$

The forced output response to an input function that satisfies $|u(t)| \leq K$ for every $t \geq 0$ has absolute value

$$|y_f(t)| = \left| \sum_{\sigma=0}^{t-1} HF^{t-1-\sigma} \mathbf{g}u(\sigma) \right| = \left| \sum_{\sigma=0}^{t-1} \left(\frac{1}{2}\right)^{t-1-\sigma} u(\sigma) \right| \leq \sum_{\sigma=0}^{\infty} \left(\frac{1}{2}\right)^{\sigma} K = 2K. \quad (8.31)$$

Therefore the forced output is bounded, too, and the system is externally stable. However, suppose that the initial state of the system is different from zero and has the value

$$\mathbf{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

In this case, the output that corresponds to a bounded input is the sum of the forced component, satisfying (8.31), and of an unforced component given by

$$y_u(t) = HF^t \mathbf{x}(0) = 2^t.$$

It is clear that the output $y(\cdot) = y_f(\cdot) + y_u(\cdot)$ cannot remain bounded as t increases, while the input sequence is so.

In this section we investigate how the external stability of a linear model $\Sigma = (F, G, H)$ is connected with

- the behavior of the unforced evolution of the state variables, and, more precisely, with the so called “*internal stability*” (i.e. the asymptotic stability discussed in Chapter 3),
- the analytic properties of the transfer matrix $W(z)$ realized by Σ .

In the subsequent discussion we will take advantage of the following lemma

Lemma 8.3.1 [POLES AND EIGENVALUES] *If $\Sigma = (F, G, H)$ is a realization of the strictly proper rational matrix $W(z)$, every pole⁵ of $W(z)$ is an eigenvalue of F .*

³As it is well-known, BIBO stands for bounded input/bounded output.

⁴We endow the spaces of input and output sequences with the ℓ_∞ norm; on the other hand the spaces $\mathbb{R}^m$ and $\mathbb{R}^p$, to which the samples $\mathbf{u}(t)$ and $\mathbf{y}(t)$ belong, are equipped with the ℓ_2 norm. Adopting different norms, however, leads to the same results.

⁵A complex number p is a pole of a rational matrix if it is a pole of at least one of its entries.

If Σ is a minimal realization of $W(z)$, then the set of poles of $W(z)$ coincides with the spectrum of F .

PROVA Whatever the $\Sigma = (F, G, H)$ that realizes the rational matrix $W(z)$, the latter can be represented as

$$W(z) = \frac{H \operatorname{adj}(zI - F)G}{\det(zI - F)}.$$

Therefore the poles of $W(z)$ belong to the zero set of the polynomial $\det(zI - F)$, namely to the spectrum of F .

We assume now that $\Sigma = (F, G, H)$ is a minimal realization, and prove that, viceversa, each eigenvalue of F is a pole of $W(z)$.

Possibly by resorting to a change of basis in the state space, it is unrestrictive to suppose that F is in Jordan canonical form. Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be the distinct eigenvalues of F and denote by P_i the block diagonal matrix that encompasses all the miniblocks of F corresponding to eigenvalue λ_i . Then the transfer matrix $W(z)$ is expressed as

$$\begin{aligned} W(z) &= [H_1 \ H_2 \ \dots \ H_k] \begin{bmatrix} zI - P_1 & & & \\ & zI - P_2 & & \\ & & \ddots & \\ & & & zI - P_k \end{bmatrix}^{-1} \begin{bmatrix} G_1 \\ G_2 \\ \vdots \\ G_k \end{bmatrix} \\ &= \sum_{i=1}^k H_i (zI - P_i)^{-1} G_i. \end{aligned} \quad (8.32)$$

Notice that, for $i = 1, 2, \dots, k$, the term $W_i(z) := H_i (zI - P_i)^{-1} G_i$ either exhibits a unique pole in λ_i (the only eigenvalue of P_i), or has no poles at all, and in this case the strictly proper rational matrix $W_i(z)$ is necessarily zero.

In the sum $\sum_{i=1}^k W_i(z)$ no cancellation is possible between the poles of two distinct elements, as distinct elements possess distinct poles. Consequently a complex number p is a pole of the matrix $W(z)$ if and only if it is a pole of some element $W_i(z)$.

If some eigenvalue λ_j of F was not a pole of $W(z)$, the element $W_j(z)$ in (8.32) would be zero and it is clear that $W(z) = \sum_{i \neq j} H_i (zI - P_i)^{-1} G_i$, would be realizable in a dimension lower than the minimal one by eliminating in (F, G, H) the block of index j . ■

- EXERCISE 8.3.1 [MINIMALITY AND EIGENVECTORS] A single input/single output system $\Sigma = (F, \mathbf{g}, H)$ is reachable and observable if and only if, for any right eigenvector $\mathbf{v}$ and any left eigenvector $\mathbf{w}^T$ of F , we have

$$H\mathbf{v} \neq 0, \quad \mathbf{w}^T \mathbf{g} \neq 0. \quad (8.33)$$

‡ *Solution* Assume that Σ is reachable and observable and $\mathbf{v}$ is a right eigenvector of F that corresponds to the eigenvalue λ . Thus we have

$$\begin{bmatrix} H \\ HF \\ \vdots \\ HF^{n-1} \end{bmatrix} \mathbf{v} = \begin{bmatrix} H \\ \lambda H \\ \vdots \\ \lambda^{n-1} H \end{bmatrix} \mathbf{v},$$

and condition $H\mathbf{v} = 0$ would contradict the observability of Σ . Analogously, if $\mathbf{w}^T$ is a left eigenvector of F , then $\mathbf{w}^T [\mathbf{g} \ F\mathbf{g} \ \dots \ F^{n-1}\mathbf{g}] = \mathbf{w}^T [\mathbf{g} \ \lambda\mathbf{g} \ \dots \ \lambda^{n-1}\mathbf{g}]$ implies the inequality $\mathbf{w}^T \mathbf{g} \neq 0$.

Viceversa, assume that (8.33) holds for all right and left eigenvectors of F . If λ belongs to $\Lambda(F)$, the right eigenspace U_λ has dimension one.

In fact we have $H\mathbf{v} \neq 0$ for any nonzero $\mathbf{v} \in U_\lambda$, and consequently the linear map $H : \mathbb{C}^n \rightarrow \mathbb{C} : \mathbf{x} \mapsto H\mathbf{x}$ satisfies the condition $U_\lambda \cap \ker H = \{\mathbf{0}\}$, which amounts to saying that U_λ and $\ker H$ are direct summands. This shows that $\dim(U_\lambda) \geq 2$ and $\dim(\ker H) = n - 1$ are incompatible in a space of dimension n and the dimension of U_λ must be one. Thus the Jordan form of F includes exactly one miniblock for each eigenvalue λ_i , $i = 1, 2, \dots, k$, F is a cyclic matrix, and there is no restriction in assuming that Σ is given by

$$F = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_k \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} \mathbf{g}_1 \\ \mathbf{g}_2 \\ \vdots \\ \mathbf{g}_k \end{bmatrix}, \quad H = [H_1 \quad H_2 \quad \dots \quad H_k],$$

where $J_i \in \mathbb{C}^{\nu_i \times \nu_i}$ is the only miniblock that corresponds to the eigenvalue λ_i .

Set now $h_1 = 1$, $h_i = 1 + \nu_1 + \dots + \nu_{i-1}$ for $i > 1$, and $k_i = \nu_1 + \nu_1 + \dots + \nu_i$. As the right and left eigenvectors that correspond to λ_i are proportional to the canonical vectors $\mathbf{e}_{h_i}$ and $\mathbf{e}_{k_i}^T$, respectively, condition (8.33) implies that the last component of the blocks $\mathbf{g}_i$ and the first component of the blocks H_i are different from zero. This entails that Σ is reachable and observable.

- EXERCISE 8.3.2 [POLES AND EIGENVECTORS] Let λ be a real eigenvalue of the single input/single output system $\Sigma = (F, \mathbf{g}, H)$ and suppose that

$$H\mathbf{v}_\lambda \neq 0, \quad \mathbf{w}_\lambda^T \mathbf{g} \neq 0, \quad (8.34)$$

holds for all right eigenvectors $\mathbf{v}_\lambda$ and left eigenvectors $\mathbf{w}_\lambda^T$ that correspond to λ . Then λ is a pole of the transfer function $H(zI - F)^{-1}\mathbf{g}$.

‡ Solution We can assume that F is in Jordan form

$$F = \begin{bmatrix} P_1 & & & \\ & P_2 & & \\ & & \ddots & \\ & & & P_k \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} \mathbf{g}_1 \\ \mathbf{g}_2 \\ \vdots \\ \mathbf{g}_k \end{bmatrix}, \quad H = [H_1 \quad H_2 \quad \dots \quad H_k],$$

where the diagonal block P_i , of dimension ν_i , encompasses all miniblocks having eigenvalue λ_i , $i = 1, 2, \dots, k$. Suppose, moreover, that λ coincides with λ_1 .

The right and left eigenvectors of F that correspond to λ_1 are obtained by completing the right and left eigenvectors $\mathbf{v}_1$ and $\mathbf{w}_1^T$ of the block P_1 with $n - \nu_1$ zero components. Hence condition (8.34) amounts to assuming that $H_1\mathbf{v}_1 \neq 0$, $\mathbf{w}_1^T \mathbf{g}_1 \neq 0$ for any right eigenvector $\mathbf{v}_1$ and any left eigenvector $\mathbf{w}_1^T$ of P_1 . The previous exercise shows that the system $\Sigma_1 = (P_1, \mathbf{g}_1, H_1)$ is reachable and observable, and consequently its transfer function has a pole in λ_1 . This is a pole of the transfer function of Σ , too.

- EXERCISE 8.3.3 [INFINITESIMAL MARKOV COEFFICIENTS] Let $w(z) = p(z)/q(z) = \sum_{i=1}^{\infty} m_i z^{-i}$ be a strictly proper rational function and suppose that the polynomials $p(z)$ and $q(z)$ are coprime. Show that the Markov coefficients m_i are infinitesimal as $i \rightarrow \infty$ if and only if all zeros of the polynomial $q(z)$ have moduli smaller than 1.

‡ Solution Let $\Sigma = (F, \mathbf{g}, H)$ be a minimal realization of $w(z)$. If the zeros of $q(z)$ have moduli smaller than 1, the same holds true for the eigenvalues of F . Hence the powers F^i are infinitesimal as $i \rightarrow +\infty$ and the Markov coefficients $m_i = HF^{i-1}\mathbf{g}$ are infinitesimal, too.

On the other hand, suppose that λ is a zero of $q(z)$, and therefore an eigenvalue of F , and assume that $|\lambda| \geq 1$.

Consider the product of the observability matrix by the matrix with infinitely many columns $\mathcal{R}_\infty = [\mathbf{g} \quad F\mathbf{g} \quad \dots \quad F^{n-1}\mathbf{g} \quad F^n\mathbf{g} \quad \dots]$

$$\mathcal{OR}_\infty = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{n-1} \end{bmatrix} [\mathbf{g} \quad F\mathbf{g} \quad \dots \quad F^{n-1}\mathbf{g} \quad F^n\mathbf{g} \quad \dots] = \begin{bmatrix} m_1 & m_2 & \dots & m_{n-1} & m_n & \dots \\ m_2 & m_3 & \dots & m_n & m_{n+1} & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ m_n & m_{n+1} & \dots & m_{2n-2} & m_{2n-1} & \dots \end{bmatrix}$$

By Exercise 8.3.1, the eigenvalue λ of F has a left eigenvector $\mathbf{w}^T$ such that $\mathbf{w}^T \mathbf{g} \neq 0$. Therefore the row vector $\hat{\mathbf{w}}^T = \mathbf{w}^T \mathcal{O}^{-1}$ satisfies

$$\begin{aligned} \hat{\mathbf{w}}^T \begin{bmatrix} m_1 & m_2 & \dots & m_{n-1} & m_n & \dots \\ m_2 & m_3 & \dots & m_n & m_{n+1} & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ m_n & m_{n+1} & \dots & m_{2n-2} & m_{2n-1} & \dots \end{bmatrix} &= \mathbf{w}^T [\mathbf{g} \quad F\mathbf{g} \quad \dots \quad F^{n-1}\mathbf{g} \quad F^n\mathbf{g} \quad \dots] \\ &= (\mathbf{w}^T \mathbf{g}) [1 \quad \lambda \quad \dots \quad \lambda^{n-1} \quad \lambda^n \quad \dots]. \end{aligned}$$

This finally gives $\hat{\mathbf{w}}^T \begin{bmatrix} m_{h+1} \\ m_{h+2} \\ \vdots \\ m_{h+n} \end{bmatrix} = (\mathbf{w}^T \mathbf{g}) \lambda^h$, $h = 0, 1, \dots$, and condition $|\lambda| \geq 1$ implies that the coefficients m_i cannot be infinitesimal.

- EXERCISE 8.3.4 [RATIONAL MATRICES WITH POLES ONLY IN THE ORIGIN] Let

$$W(z) = M_1 z^{-1} + M_2 z^{-2} + \dots + M_d z^{-d}, \quad M_d \neq 0$$

be a rational strictly proper matrix endowed with poles only in the origin, and suppose that $\Sigma = (F, G, H)$ is a minimal realization of $W(z)$. Show that F is a nilpotent matrix, with nilpotency index d , and both the reachability and observability matrices of Σ contain exactly d nonzero blocks.

‡ Solution $M_d = HF^{d-1}G \neq 0$ implies $F^{d-1} \neq 0$. On the other hand, Lemma 8.3.1 shows that F is nilpotent, and consequently its index of nilpotency must exceed $d-1$.

Denoting by n the dimension of the minimal realization Σ , from $M_{d+1} = HF^d G = 0$, $M_{d+2} = HF^{d+1}G = 0$, $M_{d+3} = HF^{d+2}G = 0, \dots$ one gets

$$HF^{d+i} [G \quad FG \quad \dots \quad F^{n-1}G] = HF^{d+i} \mathcal{R} = 0, \quad i = 0, 1, 2, \dots$$

$$\text{This implies } HF^{d+i} = 0, \quad i = 0, 1, 2, \dots, \text{ and hence } \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{n-1} \end{bmatrix} F^d = \mathcal{O} F^d = 0.$$

We therefore have $F^d = 0$, so d is the nilpotency index of F . The statement that $\mathcal{R}$ and $\mathcal{O}$ contain exactly d nonzero blocks each is obvious.

The following proposition provides two characterizations of the external stability of a discrete system Σ (and of all systems having the same transfer matrix as Σ). The first one is based on the internal stability of the reachable and observable subsystem Σ_m , while the second one depends on the location of the transfer matrix poles in the open unit disk of $\mathbb{C}$. As a consequence, a proper rational matrix $W(z)$ is called (BIBO) stable⁶ when its poles have moduli less than 1.

Proposition 8.3.2 Let $\Sigma = (F, G, H)$ be a discrete system that realizes the rational matrix $W(z)$, and denote by $\Sigma_m = (A, B, C)$ the reachable and observable subsystem of Σ or, equivalently, a minimal realization of $W(z)$. The following properties are equivalent

- i) Σ is externally stable,
- ii) Σ_m is externally stable,
- iii) Σ_m is internally stable,

⁶alternatively, a Schur matrix

iv) the poles of $W(z)$ belong to the open unit disk of $\mathbb{C}$.

PROOF The equivalence (i) $\Leftrightarrow$ (ii) is obvious, as Σ and Σ_m have the same transfer matrix and, consequently, any input induces in Σ the same forced output as in Σ_m .

To demonstrate the equivalence (iii) $\Leftrightarrow$ (iv), just observe that Σ_m is internally stable if and only if the spectrum of A is included in the open unit disk of $\mathbb{C}$, which amounts to saying, by Lemma 8.3.1, that the poles of $W(z)$ belong to the same disk.

We are therefore left with the proof of the equivalence (ii) $\Leftrightarrow$ (iii), i.e. the equivalence of the internal and the external stability of a minimal system.

(iii) $\Rightarrow$ (ii) Let $\mathbf{u}(\cdot)$ be an input that satisfies the boundedness condition

$$\|\mathbf{u}(t)\| < K, \quad \forall t \geq 0.$$

The corresponding forced output satisfies the following inequalities

$$\begin{aligned} \|\mathbf{y}_f(t)\| &= \left\| \sum_{\sigma=0}^{t-1} CA^{t-1-\sigma} B \mathbf{u}(\sigma) \right\| \leq \sum_{\sigma=0}^{t-1} \|C\| \|A^{t-1-\sigma}\| \|B\| \|\mathbf{u}(\sigma)\| \\ &\leq \|C\| \|B\| K \sum_{\sigma=0}^{t-1} \|A^{t-1-\sigma}\| = \|C\| \|B\| K \sum_{\sigma=0}^{t-1} \|A^\sigma\| \leq \|C\| \|B\| K \sum_{\sigma=0}^{\infty} \|A^\sigma\| \end{aligned} \quad (8.35)$$

Since the moduli of the eigenvalues of A are smaller than 1, the series $\sum_{\sigma=0}^{\infty} \|A^\sigma\|$ converges (see Exercise 3.5.3) and its sum is a nonnegative real number M . Therefore from (8.35) we get, for all $t \geq 0$,

$$\|\mathbf{y}_f(t)\| \leq \|C\| \|B\| MK.$$

ii) $\Rightarrow$ iii) Assuming that A has an eigenvalue λ with $|\lambda| \geq 1$, we shall construct a bounded input to Σ_m that induces an unbounded forced output.

Let $\mathbf{v}$ be an eigenvector of A that corresponds to λ . If Σ_m has dimension n , there exists an input sequence $\bar{\mathbf{u}}(0), \bar{\mathbf{u}}(1), \dots, \bar{\mathbf{u}}(n-1)$ that drives the reachable system Σ_m from $\mathbf{x}(0) = \mathbf{0}$ to $\mathbf{x}(n) = \mathbf{v}$.

Assume first $|\lambda| > 1$. Upon applying to Σ_m an identically zero input from time n onwards, the state evolution of the system for $t \geq n$ is given by

$$\mathbf{x}(t) = A^{t-n} \mathbf{v} = \lambda^{t-n} \mathbf{v}. \quad (8.36)$$

The corresponding output

$$\mathbf{y}(t) = C\mathbf{x}(t) = \lambda^{t-n} C\mathbf{v}, \quad \forall t \geq n, \quad (8.37)$$

is the unforced output induced by the state $\mathbf{v} \neq \mathbf{0}$. It cannot be identically zero, since the system is observable, and consequently in (8.37) $C\mathbf{v}$ is different from $\mathbf{0}$. Therefore the norm of $\mathbf{y}(t)$ diverges as $t \rightarrow +\infty$ and the bounded input

$$\bar{\mathbf{u}}(0), \bar{\mathbf{u}}(1), \dots, \bar{\mathbf{u}}(n-1), \mathbf{0}, \mathbf{0}, \dots \quad (8.38)$$

induces an unbounded output.

Assume now $|\lambda| = 1$. We apply

- in the time interval $[0, n-1]$ an input

$$\bar{\mathbf{u}}(0), \bar{\mathbf{u}}(1), \dots, \bar{\mathbf{u}}(n-1)$$

that drives the system from $\mathbf{x}(0) = \mathbf{0}$ to $\mathbf{x}(n) = \mathbf{v}$;

- in the time interval $[n, 2n-1]$ the input

$$\mathbf{u}(n) = \lambda^n \bar{\mathbf{u}}(0), \mathbf{u}(n+1) = \lambda^n \bar{\mathbf{u}}(1), \dots, \mathbf{u}(2n-1) = \lambda^n \bar{\mathbf{u}}(n-1),$$

that drives from $\mathbf{x}(n) = \mathbf{v}$ to $\mathbf{x}(2n) = 2\lambda^n \mathbf{v}$;

- in the time interval $[2n, 3n-1]$ the input

$$\mathbf{u}(2n) = \lambda^{2n} \bar{\mathbf{u}}(0), \mathbf{u}(2n+1) = \lambda^{2n} \bar{\mathbf{u}}(1), \dots, \mathbf{u}(3n-1) = \lambda^{2n} \bar{\mathbf{u}}(n-1),$$

that drives from $\mathbf{x}(2n) = 2\lambda^n \mathbf{v}$ to $\mathbf{x}(3n) = 3\lambda^{2n} \mathbf{v}$, and so on.

The state the system reaches at time $t = kn, k = 1, 2, \dots$, is $\mathbf{x}(kn) = k\lambda^{(k-1)n} \mathbf{v}$ and the norm of the corresponding output

$$\mathbf{y}(kn) = k\lambda^{(k-1)n} C\mathbf{v}. \quad (8.39)$$

diverges. Therefore the bounded input

$$\bar{\mathbf{u}}(0), \bar{\mathbf{u}}(1), \dots, \bar{\mathbf{u}}(n-1), \lambda^n \bar{\mathbf{u}}(0), \lambda^n \bar{\mathbf{u}}(1), \dots, \lambda^n \bar{\mathbf{u}}(n-1), \lambda^{2n} \bar{\mathbf{u}}(0), \lambda^{2n} \bar{\mathbf{u}}(1), \dots \quad (8.40)$$

induces an unbounded forced output.

If λ and $\mathbf{v}$ have complex values, the input (8.38) or (8.40) and the corresponding forced output have complex values, too. However, since the matrices of the system are real, the real and the imaginary parts of the input induce the real and the imaginary part of the output, respectively.

If the output is unbounded, it is possible that its real part is unbounded. In this case the bounded input provided by the real part of (8.38) or (8.40) induces an unbounded real output. If the output has a bounded real part, its imaginary part is necessarily unbounded. In this case, the product of the imaginary part of (8.38) or (8.40) multiplied by the imaginary unit provides a bounded real input that induces an unbounded real output. ■

Example 8.3.1 The system

$$\begin{aligned} \mathbf{x}(t+1) &= \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t) = F\mathbf{x}(t) + \mathbf{g}u(t) \\ y(t) &= [1 \quad 1] \mathbf{x}(t) = H\mathbf{x}(t) \end{aligned} \quad (8.41)$$

is a minimal realization of the rational function

$$W(z) = \frac{1}{z-1} + \frac{1}{z-1/2}.$$

The system is not externally stable, however no input of finite *duration* exists, that induces an unbounded forced output. In fact, consider any input $\mathbf{u}(\cdot)$ satisfying $\mathbf{u}(t) = \mathbf{0}$ for all $t \geq \tau \geq 0$; the output on $[\tau, +\infty)$ is induced by the state

$$\mathbf{x}(\tau) = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

hence it is unforced

$$y(t) = HF^{t-\tau} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha \left(\frac{1}{2}\right)^{t-\tau} + \beta, \quad t \geq \tau$$

and bounded, whatever α and β may be.

It is possible to design an input with bounded norm that induces an unbounded forced output, but the input duration must be infinite. Mimicking the proof of the previous proposition, we consider the eigenvector

$$\mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

that corresponds to the eigenvalue $\lambda = 1$ and the input sequence $\bar{u}(0) = 2, \bar{u}(1) = -1$ that drives the system from $\mathbf{x}(0) = \mathbf{0}$ to $\mathbf{x}(2) = \mathbf{v}$. It is easy to verify that the (periodic) input

$$u(t) = \begin{cases} -1, & \text{se } t = 1, 3, 5, \dots; \\ 2, & \text{se } t = 0, 2, 4, \dots \end{cases}$$

produces an unbounded output, with $y(2k) = k, \forall k \geq 0$.

Proposition 8.3.2 holds true also for continuous time systems, provided that the stability conditions regarding the poles of the transfer matrices and the eigenvalues of the state updating matrices refer to the left half plane $\Re(s) < 0$ instead of the unit disc $|z| < 1$ of the complex field. Next exercises provide some hints about the proof.

- EXERCISE 8.3.5 Internal stability of the continuous time system $\Sigma = (F, G, H)$ implies its external stability.

‡ Hint: (i) If $-\xi < 0$ denotes the maximum among the real parts of the eigenvalues of F , then the matrix $e^{\frac{\xi}{2}t} e^{Ft}$ is infinitesimal as $t \rightarrow +\infty$.

ii) There exists a positive real number M such that $\|e^{Ft}\| < Me^{-\frac{\xi}{2}t}$, for all $t \geq 0$.

iii) If an input satisfies the inequality $\|\mathbf{u}(t)\| < K, \forall t \geq 0$, the forced output is bounded by $\|\mathbf{y}_f(t)\| \leq \|H\| \|G\| MK \int_0^\infty e^{-\frac{\xi}{2}t} dt$.

- EXERCISE 8.3.6 Suppose that the continuous time system $\Sigma = (A, B, C)$ is a minimal realization of $W(s)$. If Σ fails to be internally stable, then it is not externally stable.

‡ Hint: Let λ be an eigenvalue of A with nonnegative real part, $\mathbf{v}$ be the corresponding eigenvector and $\bar{\mathbf{u}}(\cdot)$ an input function with bounded norm and support included in $[0, T)$, that drives the system from $\mathbf{x}(0) = \mathbf{0}$ to $\mathbf{x}(T) = \mathbf{v}$. Consider a new input function $\mathbf{u}(\cdot)$, defined as the sum of the series

$$\bar{\mathbf{u}}(\cdot) + e^{\lambda T}(\sigma^{-T}\bar{\mathbf{u}})(\cdot) + e^{2\lambda T}(\sigma^{-2T}\bar{\mathbf{u}})(\cdot) + \dots,$$

where σ^Δ is the backward shift operator of amplitude Δ . Then

(i) The input $\mathbf{u}(\cdot)$ has bounded norm and induces at time instants kT the states $ke^{(k-1)\lambda T}\mathbf{v}$.

(ii) As k diverges, the forced output is unbounded at time instants kT .

Once again we focus our interest on discrete time systems, to deepen the analysis of some properties of the linear input/output maps.

to ensure the BIBO property. Moreover, when the conditions are fulfilled, we will study what relationship exists between the energy of the input and that of the output.

8.4 Bibliography

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